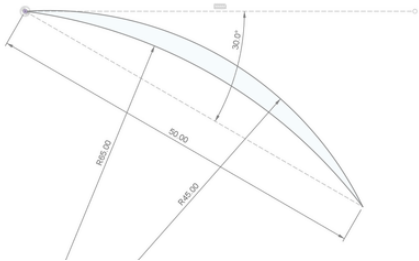
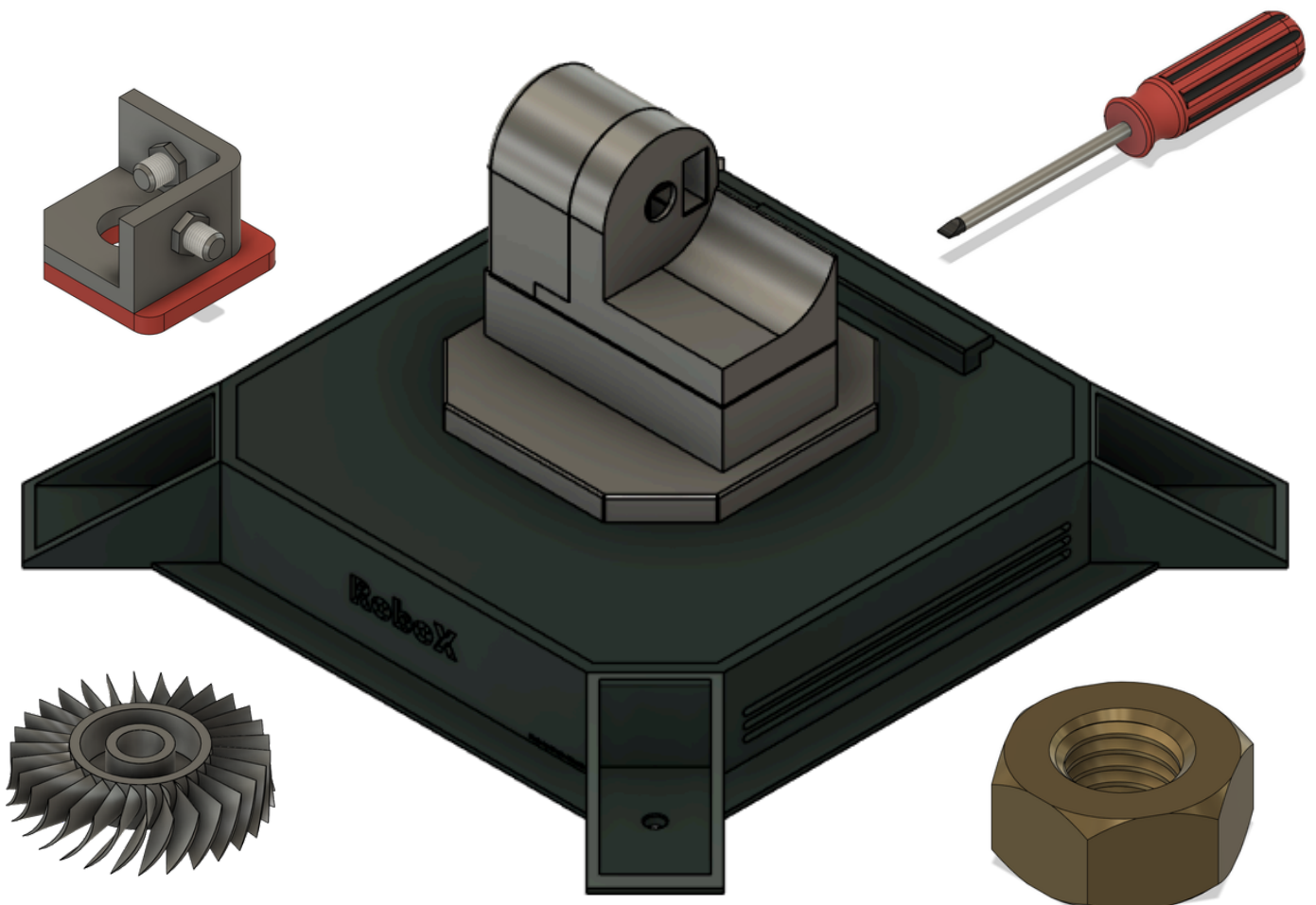


CAD & Design Portfolio

Fusion 360

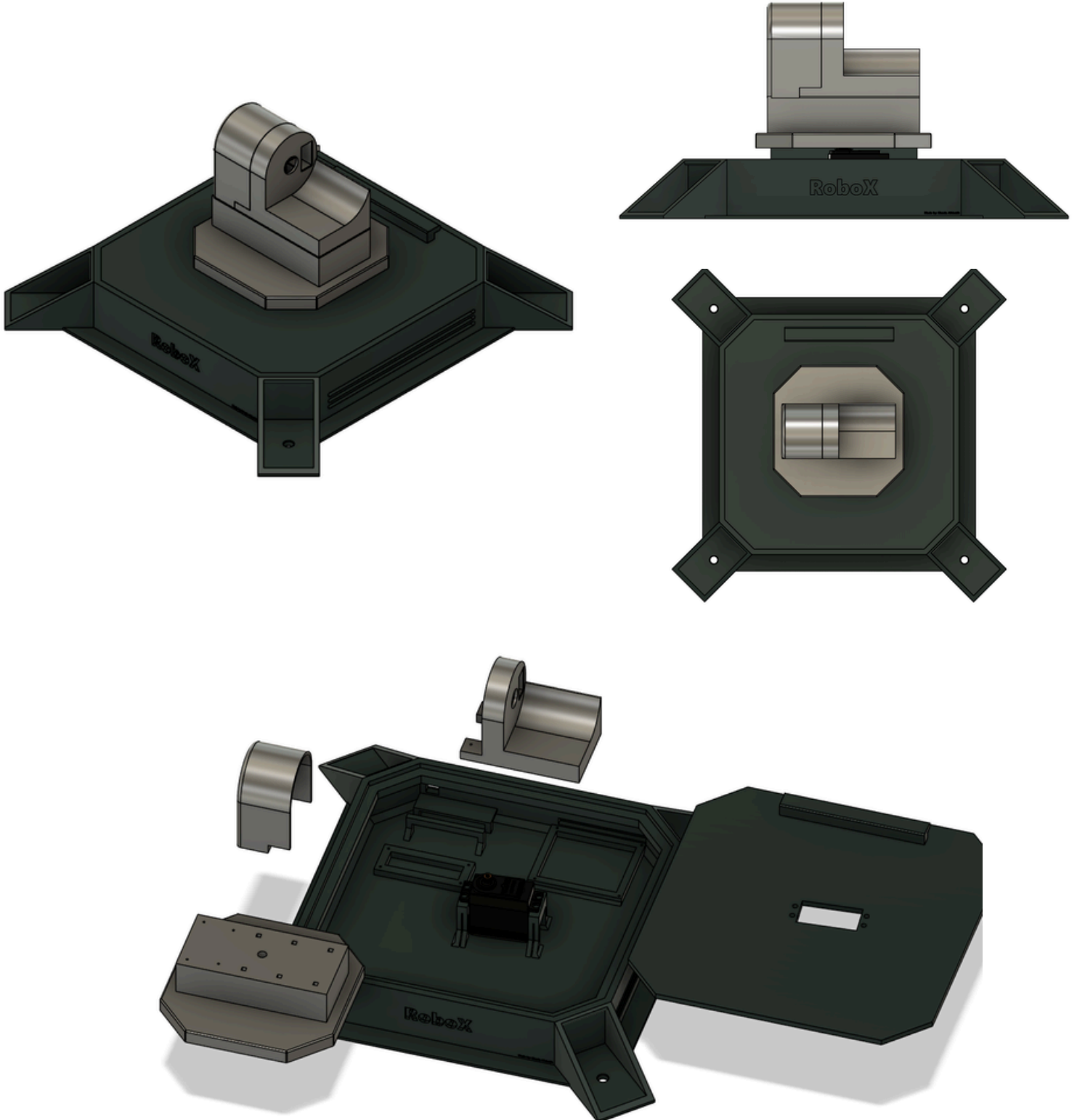


Obada Alkhatib



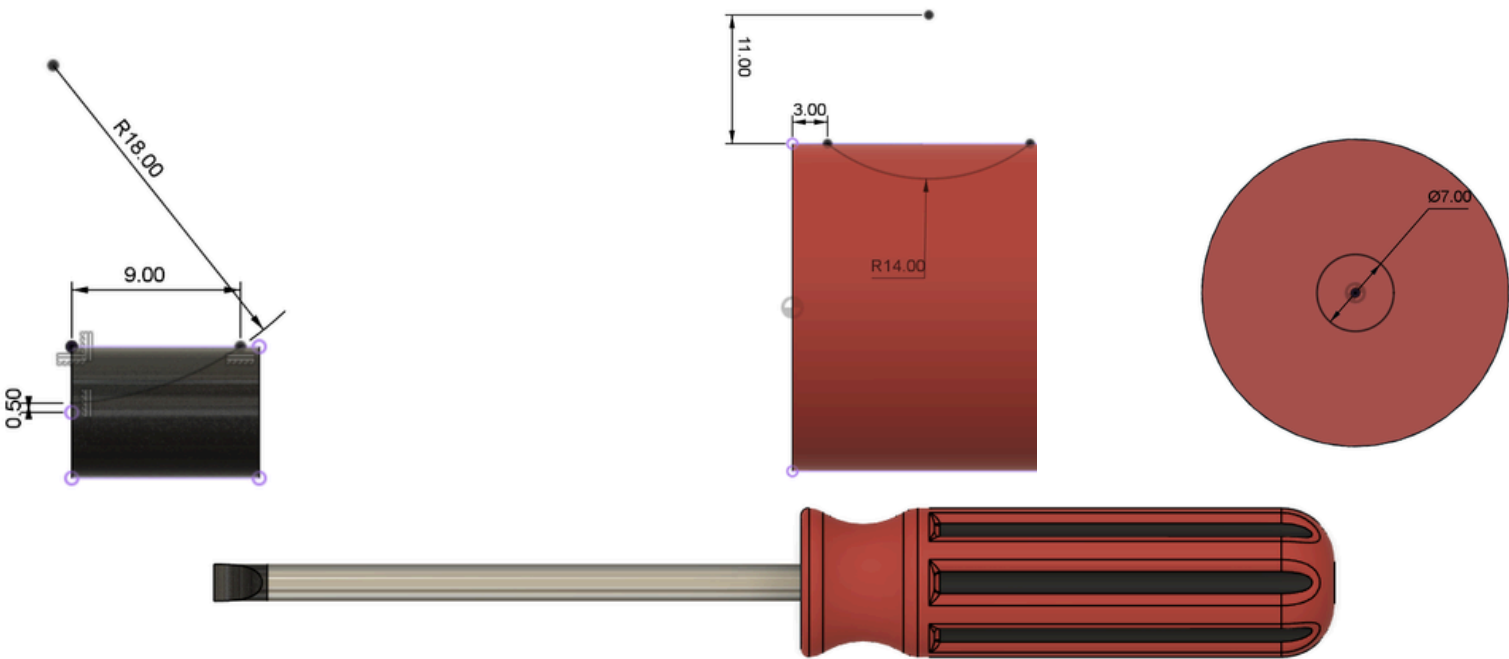
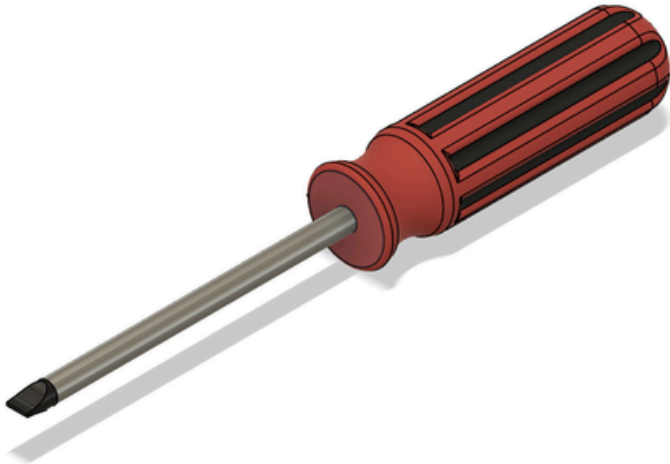
RoboX — Base & First Arm Assembly

This project is the base unit and first articulating arm of a larger robotics assembly, designed as a multi-part system for electronics housing and motion. The base was modeled as an octagonal enclosure with a removable top cover, internal mounting features, and corner gussets, providing a sealed compartment for the electrical components and servo. The first arm was built from a separate bracket and pivot housing, assembled onto the base via a keyed shaft and fastener bore to allow rotation. The exploded view highlights how the cover, bracket, mounting plate, and internal servo component come together, with the assembled result shown in front, top, and isometric views.



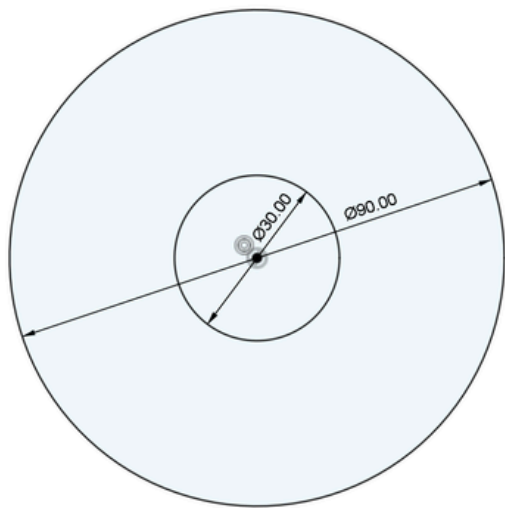
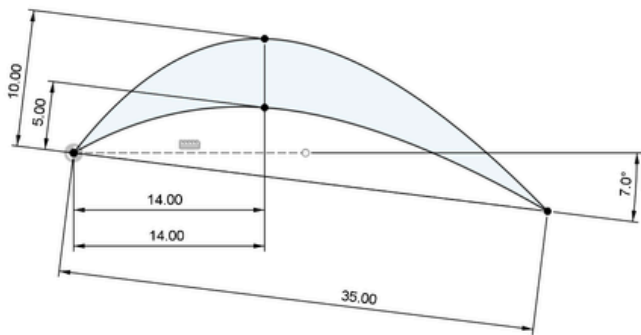
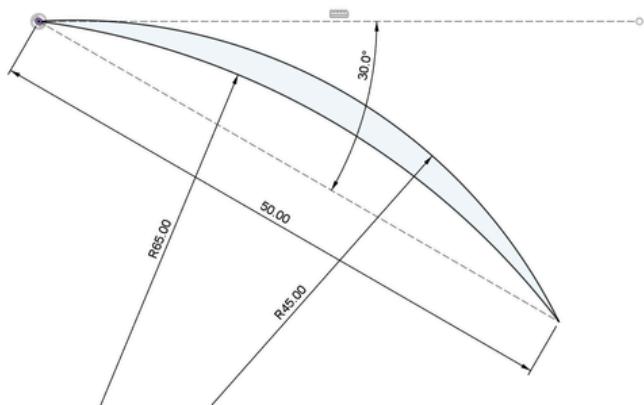
Flathead Screwdriver

This project is a fully detailed flathead screwdriver modeled in CAD, combining an ergonomic grip handle with a precisely profiled steel shaft and tip. The handle was built from a swept profile with rounded ends and ribbed finger grooves for grip, while the shaft tip was shaped with calculated radii and chamfers to form the flat blade. The final model is shown in front, top, and isometric views, demonstrating clean surfacing and proportional detail across the full assembly.



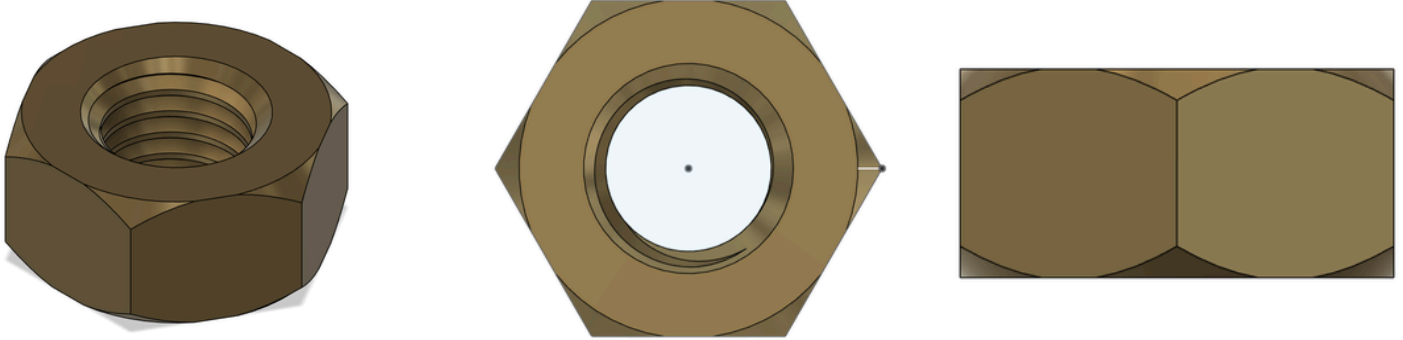
Axial Turbine Wheel

This project is a fully parametric axial turbine wheel modeled in CAD. The hub was defined with a bore-and-flange sketch, while each blade used a swept airfoil profile with controlled radii, chord length, and stagger angle to capture realistic curvature. The blade was then circularly patterned around the hub to produce the complete 24-blade wheel, shown here in front, isometric, and side views.



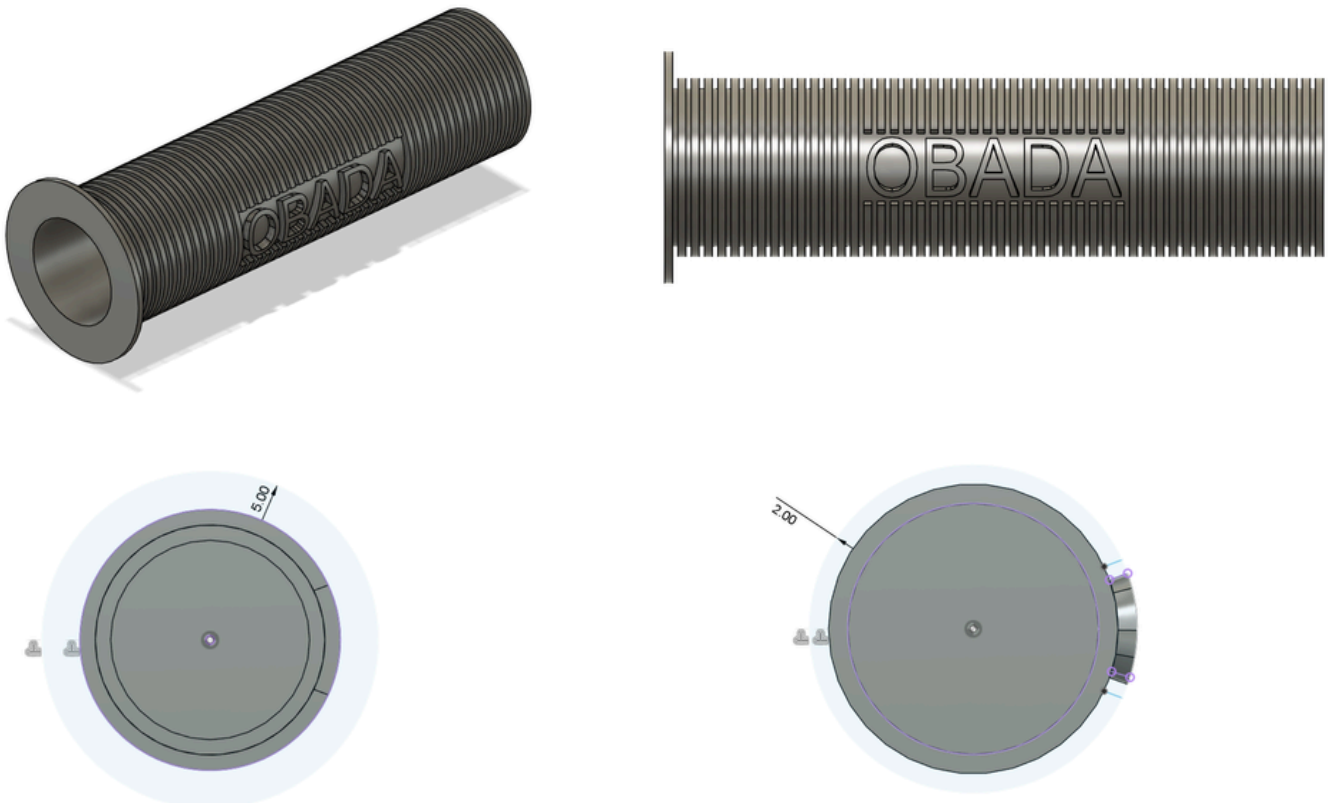
Hex Nut

This project is a fully modeled hex nut built with accurate threading and chamfered edges. The hexagonal body was sketched with a 12 mm distance-across-flats and a 1.5 mm edge chamfer, while the bore was cut and threaded to an internal $\varnothing 12$ profile. The result is a precise, standard-style fastener model shown in front, top, and isometric views.



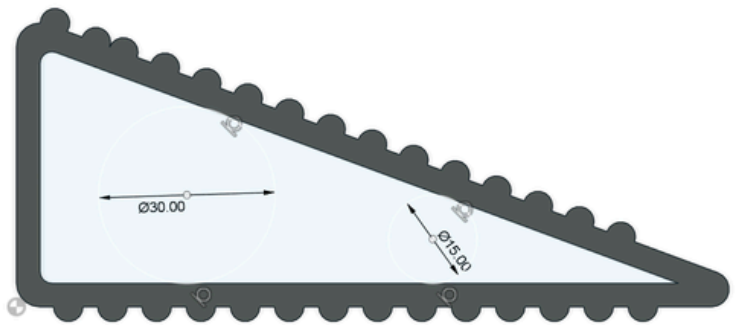
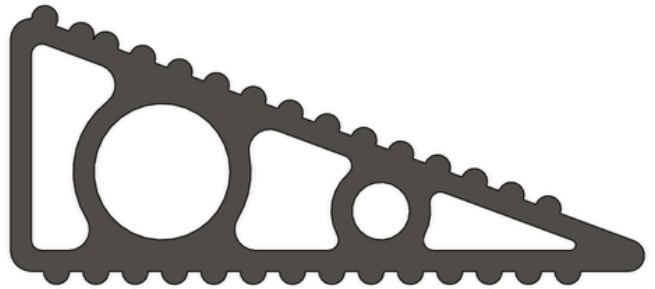
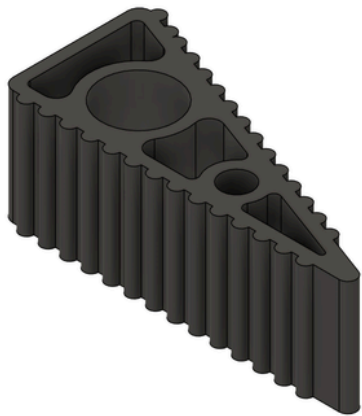
Bike Grip

This project is a textured bicycle handlebar grip modeled with a ribbed surface pattern and an embossed "OBADA" logo. The grip body was built from a $\varnothing 30$ cylindrical profile with a flared end cap, then detailed with circumferential ridges for tactile grip and a raised text feature along the shaft. The final model is shown in front, top, and isometric views, highlighting the surface detailing and overall ergonomic profile.



Door Stop

This project is a wedge-shaped rubber door stop modeled with a ribbed grip surface and lightweight internal cutouts. The wedge profile was built from a 135 x 50 mm triangular base with filleted edges, then hollowed out with circular and triangular relief cuts to reduce material while keeping structural strength. A scalloped tread pattern was added along the full perimeter for traction. The final model is shown in front, top, and isometric views.



more...

